

# ARASH JALIL KHABBAZI

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## SUMMARY

I'm a PhD candidate in Mechanical Engineering at Purdue University, working on modeling, control, optimization, and machine learning for building energy systems. My research builds practical automation for efficient, occupant- and grid-interactive buildings, focusing on predictive control and reinforcement learning. I develop and demonstrate scalable data-driven control algorithms in real buildings, and I also work on power control systems in homes and on demand flexibility in commercial buildings. My work combines rigorous methods with real-world testing to make control systems more reliable and easier to deploy at scale.

## EDUCATION

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Purdue University</b> , <i>West Lafayette, IN</i><br>PhD in Mechanical Engineering<br>Concentration in Computational Science and Engineering                                     | <i>08/2023 – Present</i><br><b>GPA: 3.9/4.0</b>                           |
| <b>University of British Columbia (UBC)</b> , <i>BC, Canada</i><br>MS in Mechanical Engineering<br><i>Thesis: "Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines"</i> | <i>05/2021 – 08/2023</i><br><b>GPA: 94% (4.0/4.0)</b>                     |
| <b>University of Tabriz</b> , <i>EA, Iran</i><br>BS in Mechanical Engineering                                                                                                       | <i>08/2016 – 07/2020</i><br><b>GPA: 19.12/20 (4.0/4.0) – Rank: 1/155+</b> |

## SELECT AWARDS AND RECOGNITION

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|-------------------------------------------------------------------------------------------------|-------------|
| <b>ASHRAE Graduate Student Grant-in-Aid Award</b>                                               | <i>2025</i> |
| <b>Best PhD Forum Presentation Runner-up Award</b> , <i>ACM BuildSys</i>                        | <i>2025</i> |
| <b>ACM SIGEnergy Student Travel Grant</b> , <i>ACM BuildSys</i>                                 | <i>2025</i> |
| <b>Best Paper Award</b> , <i>CSME International Congress</i>                                    | <i>2023</i> |
| <b>Best Presentation Award</b> , <i>CSME International Congress (Advanced Energy Symposium)</i> | <i>2022</i> |
| <b>Graduate Research Scholarship</b> , <i>University of British Columbia</i>                    | <i>2022</i> |
| <b>Graduate Dean's Entrance Scholarship</b> , <i>University of British Columbia</i>             | <i>2021</i> |

## PUBLICATIONS

### Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, "Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review." *Applied Energy*, 2025. [🔗](#)
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, "Mixing hydrogen into natural gas distribution pipeline system through Tee junctions." *International Journal of Hydrogen Energy*, 2024. [🔗](#)

### Preprints

1. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher, "Indirect Data-Driven Predictive Control and the State-Space Predictor." *arXiv:2602.10936*, 2026. *Under review*.

### Conference Papers

6. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher, "Scalable Supervisory HVAC Control for Linear Objectives." *65th IEEE Conference on Decision and Control (CDC)*, 2026.
5. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher, "Evaluating Power Control Strategies for Avoiding Residential Panel Upgrades in Cold Climates." *ACEEE Summer Study on Energy Efficiency in Buildings*, 2026.
4. **A.J. Khabbazi** and K.J. Kircher, "Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings." *American Control Conference (ACC)*, 2026.

3. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, “Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study.” *CISBAT*, 2025.
2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference*, 2024.
1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *Canadian Society for Mechanical Engineering (CSME) International Congress*, 2023. **Best Paper Award**.

## Abstracts

3. **A.J. Khabbazi**, “PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, 2025. **Best PhD Forum Presentation Runner-up Award**.
2. **A.J. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *Canadian Society for Mechanical Engineering (CSME) International Congress*, 2022. **Best Presentation Award**.
1. **A.J. Khabbazi**, R. Li, J. Quinn, and M. Hoorfar, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII)*, 2021.

## TALKS

8. “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.” *American Control Conference (ACC)*, New Orleans, LA, 2026.
7. “Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *PhD Forum, 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys)*, Golden, CO, 2025.
6. “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference*, Purdue University, West Lafayette, IN, 2024.
5. “Field Demonstrations of Advanced Commercial HVAC Control.” *Intelligent Building Operations Workshop*, Purdue University, West Lafayette, IN, 2024.
4. “Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines.” *Herrick Graduate Seminar*, Purdue University, West Lafayette, IN, 2023.
3. “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *CSME International Congress*, Université de Sherbrooke, QC, Canada, 2023.
2. “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *CSME International Congress*, University of Alberta, AB, Canada, 2022.
1. “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII)*, 2021.

## RESEARCH EXPERIENCE

**Purdue University**, Graduate Research Assistant, Kircher Research Group, West Lafayette, IN 08/2023–Present

- Building the supervisory control layer and data-driven controllers for a live field demonstration in a campus building.
- Co-developed a commissioning-free linear-programming supervisory HVAC controller (CDC’26).
- Co-developed an indirect data-driven predictive control method (TPC) built on a state-space predictor (under review).
- Derived a multi-zone model of how cross-zone heat transfer inflates measured HVAC savings (ACC’26; under review).
- Co-developed active current-limiting control for home appliances that avoids panel upgrades (CISBAT’25, ACEEE’26).
- Reviewed 100+ field demonstrations of model predictive control and reinforcement learning (Applied Energy 2025).

**UBC**, Graduate Research Assistant, TMMF Lab & FortisBC Renewable Gas, BC, Canada 05/2021–08/2023

- Ran CFD simulations of hydrogen blending into natural gas pipelines; built ideal- and real-gas mixture models in C. 
- Delivered 1 journal paper, 3 conference papers, an MS thesis, and 2 awards on a project that seeded the UBC H<sub>2</sub>Lab. 

## INDUSTRY EXPERIENCE

**Purdue Energy & Utilities**, Graduate Researcher, Energy Systems, West Lafayette, IN 02/2026–Present

- Leading real-time data pipelines and predictive models for a pre-cooling pilot shifting campus cooling demand off-peak.
- Delivered a nine-year energy-use assessment of 21 renovations to Purdue leadership for the Giant Leaps Master Plan.
- Quantified ~\$41k/yr savings and a statistically significant 8% lower approach from a two-plant air/dirt-separator retrofit.

**Tesla, Inc.**, Applied Machine Learning Intern, Drive Unit, *Sparks, NV*

*05/2025–08/2025*

- Built MySQL pipelines, engineered features, and trained ML models (XGBoost, random forests, neural nets) with  $R^2$  0.85.
- Helped deploy a thermal imaging system for drive-unit inspection, replacing destructive laboratory testing.
- Delivered ML-driven quality gains projected to cut testing costs by over \$1M/year; offered a three-month extension.

## TEACHING EXPERIENCE

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**Purdue University**, Graduate Teaching Assistant (PhD)

- ME597: Distributed Energy Resources (DERs) 🌐

*Spring'26*

- Supported 52 students on modeling and control of DERs such as batteries, solar PV, EVs, and HVAC.
- Assisted with homework, coding modules, and office hours covering DER concepts.
- Guided semester projects on DER topics, including building controls and data center cooling.

**University of British Columbia**, Graduate Teaching Assistant (MS)

- Engineering Analysis I
- Fluid Mechanics II Lab

*Fall'21, Spring'22, Fall'22*

*Spring'22, Spring'23*

## LEADERSHIP & MENTORSHIP

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**Research Mentor**, Purdue University

*2024–Present*

- Mentored three master's students and one undergraduate on research planning and technical writing.
- Collaborated on their research and co-authored the resulting publications.

**Disability Resource Centre Invigilator**, University of British Columbia

*2021–2023*

- Invigilated accommodated exams, supporting students with approved testing accommodations.

## ACADEMIC SERVICE

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**Conference and workshop organizing:**

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|-----------|--------------------------------------------------------------------------------------------|
| July 2026 | Student Chair, Intelligent Building Operations (IBO) Workshop 🌐, <i>West Lafayette, IN</i> |
| July 2024 | Student Chair, Intelligent Building Operations (IBO) Workshop, <i>West Lafayette, IN</i>   |

**Society memberships:** ASHRAE, ACM, IEEE, CSME

**Reviewing:** IEEE Transactions on Control Systems Technology; IEEE Conference on Decision and Control; Building and Environment; Journal of Building Performance Simulation; ACEEE Summer Study on Energy Efficiency in Buildings; Herrick Conferences

## SELECT COURSES

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|-------------------------------------------------|-----------------------------------------------|
| • ECE570: Artificial Intelligence               | • STAT511: Statistical Methods                |
| • IE690: Reinforcement Learning & Control (RLC) | • ME597: Industrial Internet of Things (IIoT) |
| • AAE568: Applied Optimal Control & Estimation  | • ME597: Distributed Energy Resources (DERs)  |
| • AAE 561: Intro to Convex Optimization         | • ME511: Analysis of Thermal Systems          |

## SKILLS

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|-----------------------------------|----------------------------------------------------------------------------------------------|
| <b>Languages:</b>                 | Python, MATLAB, C, SQL                                                                       |
| <b>Optimization &amp; ML:</b>     | CVXPY/CVX, TensorFlow, scikit-learn, XGBoost, Stable-Baselines3, PyTorch, Gurobi             |
| <b>Simulation &amp; Testbeds:</b> | BOPTEST, Gymnasium, OCHRE                                                                    |
| <b>Systems &amp; Tools:</b>       | Git, InfluxDB, Grafana, MySQL, Pandas, NumPy, Linux, Docker, L <sup>A</sup> T <sub>E</sub> X |